ZOO501 Final Current Solved (Update)

Exon Intron?

Both **introns** and **exons** are the genetic material. But the essential **difference between** the two is that one is made into protein and the other is not. DNA molecules are made up of small molecules that are called nucleic acids. An **exon** is a sequence of nucleic acids that are represented **in the** mRNA molecule.

How to avoid polyspermy?

o **prevent** the entry of more than one sperm into the egg. The sea urchin egg has two mechanisms to **avoid polyspermy**: a fast reaction, accomplished by an electric change in the egg plasma membrane, and a slower reaction, caused by the exocytosis of the cortical granules

Defect of neurulation?

Neural tube **defects** (NTDs) are severe congenital malformations affecting 1 in every 1000 pregnancies. 'Open' NTDs result from failure of primary **neurulation**as seen in anencephaly, myelomeningocele (open spina bifida) and craniorachischisis.

Meroblastic cleavage?

meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg. These cells form a blastoderm.

Fraternal and Identical twin?

Twins are two offspring produced by the same pregnancy. **Twins** can be either **monozygotic**('**identical**'), meaning that they develop from one zygote, which splits and forms two embryos, or dizygotic ('**fraternal**'), meaning that each **twin**develops from a separate egg and each egg is fertilized by its own sperm cell.

What is blastocyst?

Blastocyst Blastocyst: A thin-walled hollow structure in early embryonic development that contains a cluster of cells called the inner cell mass from which the embryo arises.

How acrosomal reaction initiate in sea urchin?

In the **sea urchin**, contact with egg jelly **initiates** the **acrosome reaction**, which is a calcium-mediated process. ... The **acrosomal** vesicle (green) fuses with the plasma membrane, releasing enzymes from the tip of the sperm that aid digestion of egg jelly.

How genetic material exchange in paramecium?

The micronucleus is the generative, or germline nucleus, containing the **genetic material** that is passed along from one generation to the next. **Paramecium** reproduces asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis.

What is fate of metencephalon?

The **fate** of the brain vesicles is determined by the combination of transcription factors. Signals from the secondary organizer, the anterior neural ridge and the mesencephalon–**metencephalon** boundary (isthmus), regulate expression of the transcription factors in the adjacent region and organize its **fate**.

Function of Placenta?

The **placenta** acts to provide oxygen and nutrients to the fetus, whilst removing carbon dioxide and other waste products. It metabolises a number of substances and can release metabolic products into maternal and/or fetal circulations.

What is campomelic dysplasia?

Campomelic dysplasia is a severe disorder that affects development of the skeleton, reproductive system, and other parts of the body. This condition is often life-threatening in the newborn period. The term "**campomelic**" comes from the Greek words for "bent limb."6 days ago

Name 4 level gene regulation? Level of gene regulation?

?????

Two different pathways of neural crest cell? (10)

Neural crest cells that become the pigmentsynthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly.

The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome. **(Yehi Samjh aaya mazeed PPTS chk krain).**

Mygonesis (10)

Myogenesis: The Development of Muscle Paracrine factors instruct the Myotome cells to become muscles by inducing them to synthesize the MyoD protein. Factors from the surrounding environment induce the Pax3 transcription factor. In the absence of other inhibitory transcription factors (such as those found in the sclerotome cells), Pax3 then activates the genes encoding two muscle-specific transcription factors, Myf5 and MyoD. MyoD and Myf5 belong to a family of transcription factors called the myogenic bHLH (basic helix-loop-helix) proteins (sometimes also referred to as the MyoD family). The proteins of this family all bind to similar sites on the DNA and activate muscle-specific genes. MyoD also directly activates its own gene. While Pax3 is found in several other cell types, the myogenic bHLH proteins are specific for muscle cells. Any cell making a myogenic bHLH transcription factor such as MyoD or Myf5 is committed to becoming a muscle cell. Transfection of genes encoding any of these myogenic proteins into a wide range of cultured cells converts those cells into muscle

(Internte)

Myogenesis is the formation of **muscular** tissue, particularly during embryonic**development**. **Muscle** fibers form from the fusion of myoblasts into multi-nucleated fibers called myotubes. ... In this stage, calcium ions are critical for **development**.

Primary and Secondary neurulation?

In **primary** neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.
In **secondary** neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary

Cortical granule reaction?

Cortical granules are membrane-bound structures containing proteolytic enzymes. However, whereas each sperm contains one acrosomal vesicle, each sea urchin egg contains approximately 15,000 cortical granules

Embryonic Shield?

Also called **embryonic shield**. in the early **embryo** of mammals, the flattened inner cell mass that arises at the end of the blastocyst stage and from which the**embryo** begins to differentiate.

Slow block polyspermy?

The **slow block** to **polyspermy** begins within 10 seconds of fusion of the sperm and egg plasma membranes. A compound called inositol triphosphate (IP3) causes the release of Ca++ from intracellular stores in the egg endoplasmic reticulum.

Osteogenesis?

Osteogenesis imperfecta (OI), also known as brittle bone disease, is a group of genetic disorders that mainly affect the bones. It results in bones that break easily.

Types of Cleavage?

the entire cell is divided equally. Four major holoblastic **cleavage types** can be observed in general: radial, spiral, bilateral, and rotational. Egg cells that have larger quantities of yolk undergo meroblastic **cleavage** after fertilization, in which only a portion of the zygote undergoes**cleavage**.

Response from egg to sperm?

The responses of the egg to the sperm can be divided into "early" responses, which occur within seconds of the cortical reaction, and "late" responses, which take place several minutes after fertilization begins.

Blastocoel function?
The **blastocoel** probably serves two major **functions** in frog embryos: (1) it permits cell migration during gastrulation, and (2) it prevents the cells beneath it from interacting prematurely with the cells above it.

Tunicate gastrulation?
Tunicates, like sea urchins and vertebrates, follow the deuterostome pattern of **gastrulation** in which the blastopore becomes the anus.**Tunicate gastrulation** is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm

Protozoans? (2)
is an informal term for single-celled eukaryotes, either free-living or parasitic, which feed on organic matter such as other microorganisms or organic tissues and debris.

Stereoblastula? (2)
Blastulae produced by radial cleavage have no blastocoel and are called stereoblastulae.

Neural Tube Close? (2)
The failure of the entire neural tube to close over the entire body axis is called Craniorachischisis. Collectively, neural tube defects are not rare in humans, as they are seen in about 1 in every 500 live births.

Function of Enhancer? (3)
Enhancer function underlies regulatory processes by which cells establish patterns of gene expression. Recent results suggest that **enhancers** are specified by particular chromatin marks in pluripotent cells, which may be modified later in development to alter patterns of gene expression and cell differentiation choices.

Composition of hyaline membrane in Sea Urchin? (3)
The **hyaline** layer is actually made up of two layers, an outer lamina made primarily of hyalin protein and an inner lamina composed of fibropellin proteins

Blastocoel Function? (5)
The **blastocoel** probably serves two major **functions** in frog embryos: (1) it permits cell migration during gastrulation, and (2) it prevents the cells beneath it from interacting prematurely with the cells above it.

Blastomere cleavage? (5)
Production of a multicellular organism, many cells must be produced from the single-celled zygote. This task is accomplished by cleavage, a series of consecutive cell divisions. Cells produced during cleavage are called blastomeres.

Primary and Secondary Neural tube?
The **neural** plate folds in upon itself to form the **neural tube**, which will later differentiate into the spinal cord and the brain, eventually forming the central nervous system. Different portions of the **neural tube** form by two different processes, called **primary and secondary** neurulation, in different species.

What is Blastocyst?
A mammalian blastula in which some differentiation of cells has occurred.

How acrosomal reaction initiate in Sea Urchin?
The acrosomal reaction in sea urchins is initiated by contact of the sperm with the egg jelly. Contact with egg jelly causes the exocytosis of the sperm's acrosomal vesicle and the release of proteolytic enzymes that can digest a path through the jelly coat to the egg surface.

What is Fete metacephone?
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What is meroblastic Cleavage?
Depending mostly on the amount of yolk in the egg,**cleavage** can be holoblastic (complete or total) or**meroblastic** (partial). In **meroblastic cleavage**, only a portion of the egg divides. This is because of the large amount of yolk in the egg cell that the **cleavage**furrows do not penetrate the yolk. ... superficial**cleavage**.

Write 2 major difference pathway of nueral crest cells?

The trunk neural crest, whose cells take one of two major pathways.

Neural crest cells that become the pigmentsynthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly. The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome. Sclerotomes are blocks of mesodermal cells, derived from somites, that will differentiate into the vertebral cartilage of the spine.

The vagal and sacral neural crest, whose cells generate the parasympathetic (enteric) ganglia of the gut. The vagal (neck) neural crest lies opposite chick somites 1-7, while the sacral neural crest lies posterior to somite 28.

The cardiac neural crest is located between the cranial and trunk neural crests. The cardiac neural crest cells can develop into melanocytes, neurons, cartilage, and connective tissue (of the third, fourth, and sixth pharyngeal arches).

Cortical granule reaction? Cortical reaction?

The cortical reaction is a process initiated during fertilization by the release of cortical granules from the egg, which prevents polyspermy, the fusion of multiple sperm with one egg.

Cleavage pattern in sea urchin, nematode and tunicate?

In all four invertebrates described here, **cleavage** is holoblastic. In the **sea urchin**,**cleavage** is radial; in the snail, spiral; in the **tunicate**, bilateral; and in the **nematode**, rotational. ... In some, such as the **sea urchin** and **tunicate**, the axes are established at fertilization through determinants in the egg cytoplasm.

How secondary hypoblast form in avians?

Gastrulation of the Avian Embryo Shortly thereafter, a sheet of cells from the posterior margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the polyinvagination islands, thereby forming the secondary hypoblast.

Difference in cotice and nuclei?

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Larval Stage of Frog?

A tadpole (also called a pollywog) is the **larval stage**in the life cycle of an amphibian, particularly that of a **frog** or toad. They are usually wholly aquatic, though some species have tadpoles that are terrestrial.

Commitment Strategy?

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What happened neural tube fail to fuse?

Neural tube defects—Spina bifida, Anencephaly and Encephalocele. ... The **neural tube** goes on to develop into your baby's brain and spinal cord. When the **neural tube** does not **fuse** together properly a problem with the development of the spine or brain occurs. This is known as a **neural tube** defect.

Resting potential membrane?

The opening of the calcium channels involve the receptor's activating a cation channel (for sodium, potassium, or calcium), which would change the resting potential of the sperm plasma membrane. The calcium channels in the membrane would be sensitive to this change in membrane potential, allowing calcium to enter the sperm.

Neural plate?

The **neural plate** is a key developmental structure that serves as the basis for the nervous system. Opposite the primitive streak in the embryo, ectodermal tissue thickens and flattens to become the **neural plate**. The region anterior to the primitive knot can be generally referred to as the **neural plate**.

Difference between Cytokinesis and Karyokinesis?

Karyokinesis is the process of division of the nucleus of a cell during the phase of division (mitosis or meiosis). **Cytokinesis** is the process of division of cytoplasm of the cell. **Cytokinesis** divides cytoplasm and organelles including nuclei into its daughter cells.

Event of Fertilization?

The result of **fertilization** is a cell (zygote) capable of undergoing cell division to form a new individual. The fusion of two gametes initiates several reactions in the egg. One of these causes a change in the egg membrane(s), so that the attachment of and penetration by more than one spermatozoon cannot occur.

How Discovered sperm?

A.V Leeuwenhoek, the Dutch microscopist who co-discovered sperm in 1678, first believed them to be parasitic animals living within the semen (hence the term spermatozoa, meaning "sperm animals").

Conditional SP with example?

Conditional specification: o Characteristic of all vertebrates and few invertebrates. o Specification by interactions between cells. Relative positions are important

Tunicates gastrulation?

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm. About 4 5 hours after fertilization, the vegetal (endoderm) cells assume a wedge shape, expanding their apical margins and contracting near their vegetal margins.

Ectoderm difference?

the **ectoderm** differentiates to form the nervous system (spine, peripheral nerves and brain), tooth enamel and the epidermis (the outer part of integument). It also forms the lining of mouth, anus, nostrils, sweat glands, hair and nails

MPF Subunit?

Mitosis-promoting factor contains two subunits. υ The large subunit is called cyclin B. υ Small subunit of MPF, the cyclin-dependent kinase

Defect of Neutralization? (10)

Different neural tube defects are caused when various parts of the neural tube fail to close. Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed. Failure to close the anterior neural tube regions results in a lethal condition, anencephaly. Here, the forebrain remains in contact with the amniotic fluid and subsequently degenerates. Fetal forebrain development ceases, and the vault of the skull fails to form.

Identical Twin and Faternal Twin?

To form **identical** or monozygotic **twins**, one fertilised egg (ovum) splits and develops into two babies with exactly **the same** genetic information. To form**fraternal** or dizygotic **twins**, two eggs (ova) are fertilised by two sperm and produce two genetically unique children.

Difference mono and Polyspermy?

is that monospermy is fertilization of an oocyte by a single spermatozoon while**polyspermy** is the penetration of an ovum by more than one sperm.

What is Syncytiotrophoblast?

The **syncytiotrophoblast** is the primary structure that determines which substances cross the placenta (e.g., nutrients and oxygen) and which substances do not (e.g., maternal hormones and certain toxins).

Anatomical Approach?

The third **anatomical approach** to developmental biology is teratology, the study of birth defects. ... The fourth **anatomical approach** is mathematical modeling, which seeks to describe developmental phenomena in terms of equations.

Mantal and Ventricular zone?

new layer is called the **mantle** (or intermediate) zone, and the germinal epithelium is now called the **ventricular** zone (and, later, the ependyma).

Species Specific recognition in sea urchin?

"**Species specific recognition** of **sea urchin** gametes occurs at the levels of sperm attraction, sperm activation, and sperm adhesion to the egg surface." The **sea-urchin**egg jelly contains a **species specific** chemo-attractant (redact).

How avoid polyspermy?

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Slow block of **polyspermy**

Nevertheless, **polyspermy preventing** mechanisms are essential in mammals; a secretion reaction, the "cortical reaction" modifies the extracellular coat of the egg (the zona pellucida), and additional mechanisms that are not well understood modify the egg's plasma membrane **(Internet)**

Chemotoxix?

Is the movement of an organism in response to a chemical

Type of Egg in insect?

These are megalecithal polylecithal & centrolecithal **eggs**. The cytoplasm of **insect egg** is limited to periphery only whole central place is filled by yolk. Two **types of egg**envelops are present on **egg** of **insects**. Inner - Vitelline membrane - Primary **egg**membrane.

Two way from neural tube?

There are two major ways of forming a neural tube ¬ Primary neurulation ¬ Secondary neurulation.

Name of cell movement during gastrulation in vertebrates?

During internalization, endoderm and mesoderm **cells** move via the blastopore beneath the ectoderm. ... Patterns of **gastrulation cell movements** relative to the blastopore and the organizer are similar from **fish** to **mammals**, and conserved molecular pathways mediate **gastrulation movements.**

Vegetal pole and Animal pole?

The yolk-rich pole is referred to as the **vegetal** pole; the yolk concentration in the **animal** pole is relatively low.

Campomelic dysplasia?

Campomelic dysplasia (CMD) is a rare genetic disorder characterized by bowing of the long bones and many other skeletal and extraskeletal features. It is frequently lethal in the neonatal period due to respiratory insufficiency, but the severity of the disease is variable, and some patients survive into adulthood.

How egg protected from sunlight?

Many eggs contain ultraviolet filters and DNA repair enzymes that protect them from sunlight

Amniotic Fluid?

The amniotic fluid is the protective liquid contained by the amniotic sac of a gravid amniote. This fluid serves as a cushion for the growing fetus, but also serves to facilitate the exchange of nutrients, water, and biochemical products between mother and fetus.

What are MFN?

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What create dramatic change in the structure of organism?

Allometric growth can create dramatic changes in the structure of organisms.

Why tunicate give name?

The **name**, "**tunicate**" comes from the firm, but flexible body covering, called a tunic. Most **tunicates** live with the posterior, or lower end of the barrel attached firmly to a fixed object, and have two openings, or siphons, projecting from the other. **Tunicates**are plankton feeders.

Transcription factor group together?
General **transcription factors** are involved in the formation of a preinitiation complex. The most common are abbreviated as TFIIA, TFIIB, TFIID, TFIIE, TFIIF, and TFIIH. They are ubiquitous and interact with the core promoter region surrounding the **transcription** start site(s) of all class II genes

Difference between Enhancer and Promoter?
An **enhancer** is a sequence of DNA that functions to enhance transcription. A **promoter** is a sequence of DNA that initiates the process of transcription. A **promoter** has to be close to the gene that is being transcribed while an **enhancer** does not need to be close to the gene of interest.

Polyspermy avoided in sea urchin?
The most common way is to prevent the entry of more than one sperm into the egg. The **sea urchin** egg has two mechanisms to **avoid polyspermy**: a fast reaction, accomplished by an electric change in the egg plasma membrane, and a slower reaction, caused by the exocytosis of the cortical granules

Zona pellucida?
The **zona pellucida** supports communication between oocytes and follicle cells during oogenesis; protects oocytes, eggs, and embryos during development, and regulates interactions between ovulated eggs and free-swimming sperm during and following fertilization.

When Capacitation occur?
The set of physiological changes that allow the sperm to be competent to fertilize the egg is called capacitation. **(PPT)**

(Internet)
Capacitation is the penultimate step in the maturation of mammalian spermatozoa and is required to render them competent to fertilize an oocyte. ... In vivo,**capacitation occurs** after ejaculation, when the spermatozoa leave the vagina and enter the superior female reproductive tract.

What are protostome?
(Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location.

Composition of hyaline membrane sea urchin?
The **sea urchin** embryo **hyaline** layer is an extracellular investment which develops within 20 min postinsemination of Strongylocentrotus purpuratus eggs and contains a single calcium-precipitable subunit termed hyalin.

How neural groove formed in primary neurulation?
The process begins when the notochord induces the **formation** of the central nervous system (CNS) by signaling the ectoderm germ layer above it to **form** the thick and flat **neural plate**. ... In **primary neurulation**, the **neural plate** creases inward until the edges come in contact and fuse.

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